

[DC] Limb Motion Guidance in Extended Reality

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ABSTRACT

Virtual and augmented reality technology grants new possibilities to the movement tutorial. On the one hand, VR/AR visualize the movement instructions in the 3D world, avoiding misperceptions caused by the missing angle and depth information when using traditional 2D screens. With the tracking device in the VR/AR system, on the other hand, the users receive precise real-time feedback and even correction guidance during training to improve their performance. So far, we have run a series of user studies to propose the design implications for MR-based motion guidance in terms of user perspective, visual encoding, and motion features. And I would like to continue my research on 1) memorability of motion guidance and 2) realistic user scenarios like physiotherapy.

Index Terms: Human-centered computing—Visualization—Visualization design and evaluation methods
Computer Graphics—Three-Dimensional Graphics and Realism—Virtual Reality

1 INTRODUCTION

Effective and precise motion guidance is needed when people try to learn and practice motor skills. Traditionally, such guidance should be given by a physiotherapist or trainer, who can observe the movement performance and offer verbal or physical advice and correction in real time. However, in many situations, an external advisor might not be available for reasons like a long distance or pandemic, which means that the exercises have to be practiced alone and unsupervised. However, without real-time feedback and correction, the motions might be executed sloppily or even incorrectly, potentially leading to worsening health conditions or hard-wiring of bad habits. A widely used approach to overcome this problem is to use remote live streaming or prerecorded video tutorials for motion guidance. However, precisely following instructions provided by video tutorials on a 2D screen can be difficult. Motion paths, velocity, and postures need to be inferred from 2D screens and translated into 3D motions, which depends very much on users' perception of depth and angles. This task is challenging even if multiple perspectives are given in the video.

In this paper, we want to focus on using VR and AR technology for motion guidance. First, VR and AR technology display instructions in 3D space and allow users to change their viewpoints. Therefore, the users can view the motion instructions in 3D and receive a clear understanding of angle and depth. Second, the tracking device in the XR system can perceive the situation in the environment, so the one who makes a mistake during training will receive feedback and correction in real time.

However, how can we design the XR-based motion guidance system, which factors can affect the user's performance and memorability about motions, and how can we apply this to authentic use cases, are under-researched in this area. We attempt to fill this gap by proposing a research agenda to achieve the following research goals

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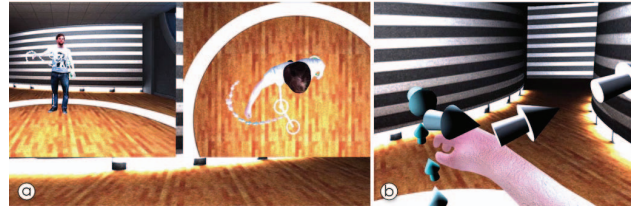


Figure 1: the setup of Study 1, with (a) 2D-mirror-person perspective (b) first-person perspective

(RG): **RG1:** Figuring out the design implications for XR-based motion guidance system; **RG2:** Evaluating the long-term effect of motion guidance on users' memory; **RG3:** Designing and evaluating the authentic unsupervised physiotherapy user scenario.

2 PROJECT FRAME

2.1 RG1 - Design Implications

While the XR-based motion guidance systems provide interesting point solutions in a large design space, a systematic understanding of the role of different design factors in the space is still lacking. For instance, no conclusive answers are available for the viewing perspective in which such a system should operate, how guidance should be visually encoded, and how different motion types influence the design. To fill this gap, we contribute three controlled user studies [3].

2.1.1 Study 1: First-Person vs. 2D Mirror

We developed a prototype, in which we visualized the movement path with the arrows and the posture with stick figures. And we used the blue rubber bands to guide the users to align their arms to the correct positions in 3D space. In the first study, we compared the first-person (Fig. 1(a)) and the 2D mirror-person perspective (Fig. 1(b)) proposed in [2].

2.1.2 Study 2: First-Person vs. Third-Person

In the second study, where we slightly modified the former prototype, we compared the first-person perspective with the third-person. Here we visualized the third-person perspective in the way of video games, in which the avatar was scaled down and put before the user. Here we designed two motions in need of both arms: one happens in the frontal area of the user, while the other in the peripheral (Fig. 2).

2.1.3 Study 3: Visual Encoding

In Study 3, we tried to figure out the effect of perspective and visual encoding on the motions with different characteristics. With a more focused visualization and motion range, we sought to replace the realistic avatar with the abstract arm token in this prototype. And here, we designed two different visual encodings: steamer, as the continuous guidance seen in Fig. 3 (a), and GhostArm as the discrete one in Fig. 3 (b). In this study, we compared the first-person (Fig. 3 (a)) with a 3D-Mirror-person perspective (Fig. 3 (c)), in which there is an additional arm token conducted a mirror-symmetric movement with the user.

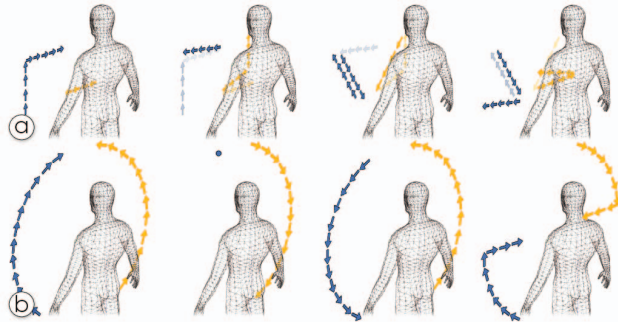


Figure 2: the motions in Study 2, (a) frontal motion (b) peripheral motion

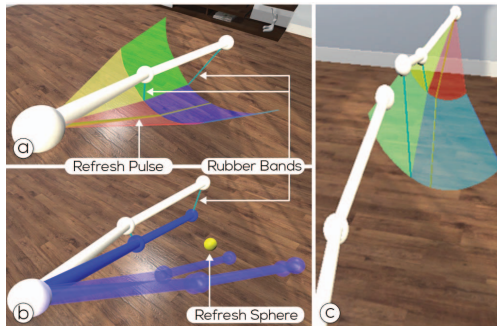


Figure 3: the setup of Study 3, with (a) continuous guidance in first-person (b) discrete guidance in first-person (c) continuous guidance in 3D-mirror-person

2.1.4 Design Implications

In total, we recruited 41 (12+12+17) participants in our studies. And the result showed that the first-person perspective is the best choice for all frontal motions. For peripheral movements, the first-person is slightly inferior to the third-person perspective in terms of completion time due to the need for head rotation to determine the position and progress of the two arms separately.

As for motion complexity, we summarized four different features of 3D movements that may affect the performance: 1) trajectory length, 2) number of intersections in a trajectory, 3) number of joints involved, and 4) number of 2D segments. Among these four features, the number of 2D segments matters the most for motion complexity.

For visual encoding in the motion guidance system, the continuity of movement instructions also matters but not as much as perspective. And the explicit visualization of the desired posture was more beneficial to maintain the postures compared to the continuity of instructions. And the continuous guidance might evoke visual clutter and prevent posture adjustment.

2.2 RG2 - Short-Term Retention

We found design implications that help users do better in following movement instructions during training. However, the precise visualization of movement instructions may not be the best way to learn a new motor skill, as they might make users too dependent and lead to poor performance when there is no guidance. My second RG is to figure out how to improve the memorability of motion guidance in XR. In this work, we would like to explore several possible approaches to improve users' short-term retention of motor skills. Concretely, we will ask the participants to follow the motion guidance together with one of the assistive techniques below several times as training, and then do the last repetition without any

instructions as an examination.

1) Post Analysis. When the participants follow the movement guidance, they will generate the movement records of their own repetitions. We visualize their movement records in the post-analysis, indicating the movement error's position, direction, and distance. So the participants may improve their performance and retention of the movement.

2) Increasing transparency. With the training repetitions increasing, we will increase the transparency of movement instructions until they are entirely invisible. This is equivalent to increasing the difficulty of the training phase, and making the subject gradually adapt to the situation without movement instructions.

3) Multimodal Feedback. We want to introduce additional feedback to magnify the participant's error during training, like the audio alarm and haptic feedback [1].

2.3 RG3 - Authentic Physiotherapy Motion Guidance

The existing motion guidance systems do not consider the actual status of the user's muscles, which is unfavorable for sports rehabilitation or muscle training, because even if the postures are similar, compensatory movement may occur. In this work, we introduce the muscle activation level as a metric in the existing motion guidance systems (and turn the simple motion-following task into a multi-objective task), where attention to both postural accuracy and muscle activation requirements is needed. The potential contribution of such a system could be: (1) Ensuring the PROPER involvement of required muscles; (2) Avoiding the involvement of unneeded muscles (compensatory movement). But how can we visualize the muscle activation levels and their requirements without introducing too much visual clutter? How should we set the boundary of both positional and muscle activation requirements?

3 CONCLUSION

XR-based motion guidance carries the potential to take the place of the human physical trainer and enable unsupervised practice. To achieve this goal, we need to figure out how to design XR-based motion guidance to ensure a good training effect and motion retention, and search for evidence in authentic user scenarios like physiotherapy. I want to outline some of the questions that could be discussed during the consortium that will better shape the work: (1) What other design parameters could be included to improve the memorability of 3D movement tasks that I have not thought about? Any psychological knowledge behind these factors? (2) How should I concretely set up the user study of memorability, especially **technically** measuring memorability? Will it be the similarity of 3D trajectories? (3) Any questionnaire about visual clutter (in VR)? (4) Any factors I miss for the XR-based physiotherapy practicing system? or any other suggestions?

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